Subject: Elements of Robotics (OE-4)								
Program: B. Tech. (All Branches; Except Mech.)				Subject Code: ME0424			Semester: IV	
Teaching Scheme				Examination Evaluation Scheme				
Lecture	Tutorial	Practical	Credits	University Theory Examination	University Practical Examination	Continuous Internal Evaluation (CIE) Theory	Continuous Internal Evaluation (CIE) Practical	Total Marks
3	0	0	3	16/40	NA	24/60	NA	100

Course objectives:

1. This course aims to familiarize students with basic terminologies of the robotics and essential knowledge to be acquainted in the field of Robotics.
2. It also aims to inculcate thorough understanding about basic terminologies, grippers, sensors, actuators and robot kinematics.

Course Outcomes (CO):

1. Understand terminologies related to robotics
2. Identify gripper, sensor and actuator of a robot
3. Apply mathematics for manipulator positioning and motion planning
4. Analyze robot mechanism using kinematics
5. Acquainted with various applications and futuristic robotic technology

Course Contents:

Unit	Contents	Hours
1	**Introduction to robotics** Brief History, Definition, Robot Anatomy, Three laws, Classification of robots, Robot terminologies: work volume, Degree of Freedom, resolution, accuracy, repeatability, dexterity, compliance, payload capacity, speed of response etc., Wrist assembly, Joint notations, Selection criteria of any robot, Industrial applications of robot, Futuristic robotics.	5
2	**Robot drive systems, End effectors and Automation** Types of drives – Hydraulic, Pneumatic and Electric, Comparison of all such drives, DC servo motors, Stepper motors, AC servo motor – salient features and applications, pulse count calculations, End effectors - Types of Grippers – Mechanical, Magnetic, vacuum, pneumatic and hydraulic, selection and design considerations,	7
3	**Robot sensors** Need for sensors, types of sensors used in Robotics, classification and applications of sensors, Characteristics of sensing devices, Selections of sensors.	3

4	**Machine Vision** Robot Vision setup (RVS), block diagram, components, working of RVS, Human vision Vs Robot Vision, Gradient calculations, Applications of RVS	**3**

Reference Books:

1. S. K. Saha, Introduction to Robotics 2e, TATA McGraw Hills Education (2014)
2. AsitavaGhoshal, Robotics: Fundamental concepts and analysis, Oxford University Press (2006)
3. Dilip Kumar Pratihar, Fundamentals of Robotics, Narosa Publishing House, (2019)
4. R. K. Mittal, I. J. Nagrath, Robotics and Control, TATA McGraw Hill Publishing Co Ltd, New Delhi (2003)
5. S. B. Niku, Introduction to Robotics – Analysis, Contro, Applications, 3rd edition, John Wiley & Sons Ltd., (2020)
6. J. Angeles, Fundamentals of Robotic Mechanical Systems Theory Methods and Algorithms, Springer (1997)
7. MikellGroover, Mitchell Weiss, Roger N. Nagel, Nicholas Odrey, AshishDutta, Industrial Robotics 2nd edition, SIE, McGraw Hill Education (India) Pvt Ltd (2012)
8. R. D. Klafter, Thomas A. Chmielewski, and MechaelNegin, Robotic Engineering – An Integrated Approach, EEE, Prentice Hall India, Pearson Education Inc. (2009)
9. J. J. Craig, “Introduction to Robotics: Mechanics and Control”, 3rd edition, AddisonWesley (2003)